Sensores capacitivos

$$Q = C \cdot U$$

$$C = \varepsilon_r \varepsilon_0 \frac{A}{d}$$

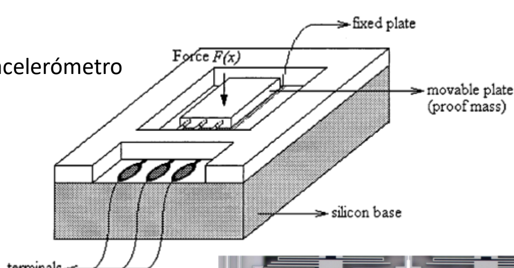
Como se pode observar, pode-se intervir em qq de uma das três componentes

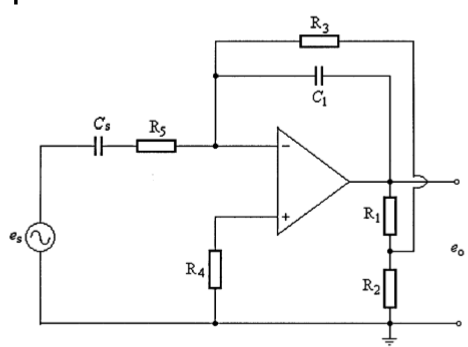
ε_r, A, d

1

Sensores capacitivos

Um acelerómetro





Um amplificador de Carga

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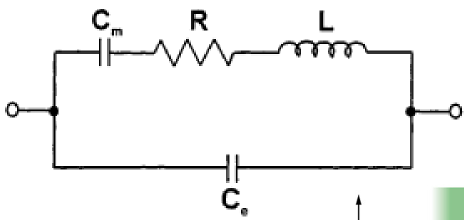
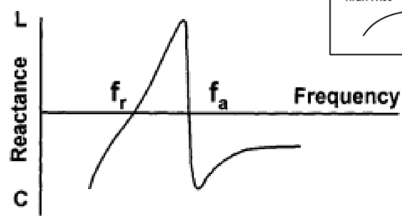
2

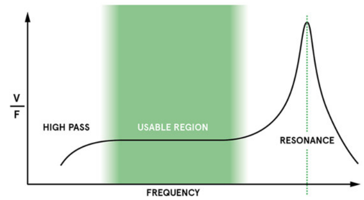
Piezoelectricidade

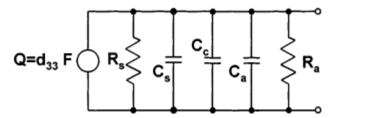


a)

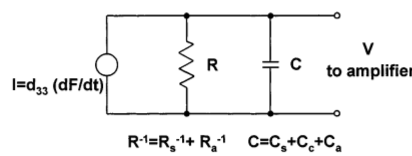


b)



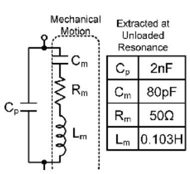


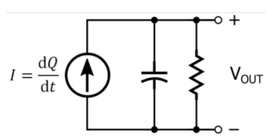
a)

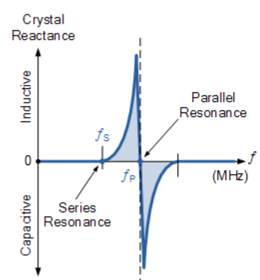


b)

Piezoelectricidade







$$X_S = R^2 + \left(X_{LS} - X_{CS}\right)^2$$

$$X_{CP} = \frac{-1}{2\pi f C_P}$$

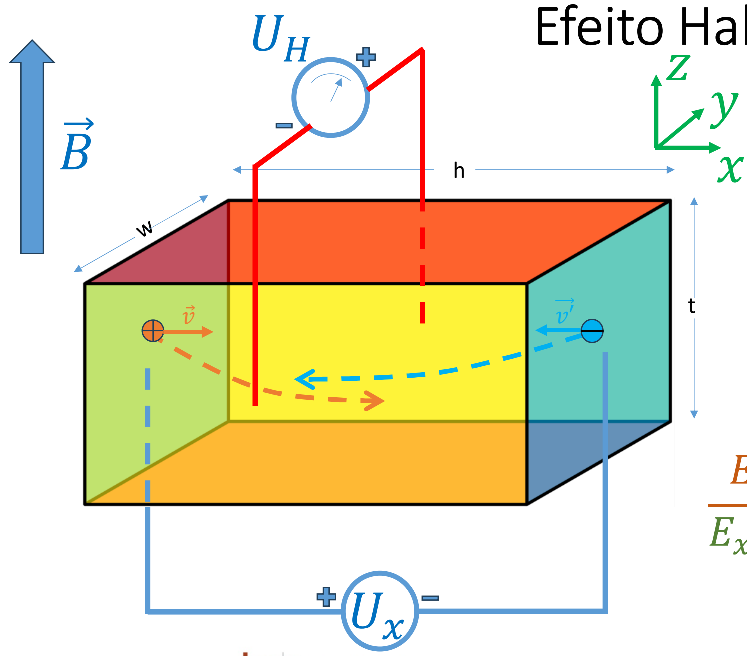
$$X_P = \frac{X_S \times X_{CP}}{X_S + X_{CP}}$$

$$f_S = \frac{1}{2\pi\sqrt{L_S C_S}}$$ Série

$$f_P = \frac{1}{2\pi\sqrt{L_S\left(\frac{C_P C_S}{C_P + C_S}\right)}}$$ Paralelo

$$Q = \frac{X_L}{R_m} = \frac{2\pi f_S L_m}{R_m}$$

Efeito Hall



$$\vec{F} = q\left(\vec{E} + \vec{v} \otimes \vec{B}\right)$$

$$J_x = \frac{I_x}{wt} = J_{nx} + J_{px}$$
$$= q\left(\mu_n n + \mu_p p\right)E_x$$

$$J_y = 0 = J_{ny} + J_{py}$$
$$= q\mu_n n\left(E_y + \mu_n E_x B_z\right) +$$
$$q\mu_p p\left(E_y - \mu_p E_x B_z\right)$$

$$\frac{E_y}{E_x B_z} = \cdots$$ Mas como $$E_x = \frac{J_x}{(\ldots)}$$

5

Efeito Hall

Então, define-se **Constante de Hall** como (S. Internacional m^3C^{-1}
ou unidades **inversas** seja de *densidade de carga* seja de *carga volúmica*)

$$R_H = \frac{E_y}{J_x B_z} = \frac{p\mu_p^2 - n\mu_n^2}{q\left(p\mu_p + n\mu_n\right)^2} \approx \frac{1}{n \cdot e}$$

$$U_H = \frac{I_x B_z}{n \cdot e \cdot t} = \frac{R_H I_x B_z}{t}$$

6

Sensores indutivos

$$-L \cdot \frac{di}{dt} = U \qquad L = K_f N^2 \mu$$

Aqui, já não é tão fácil intervir pq o factor de forma depende tb de N (Nº de espiras)

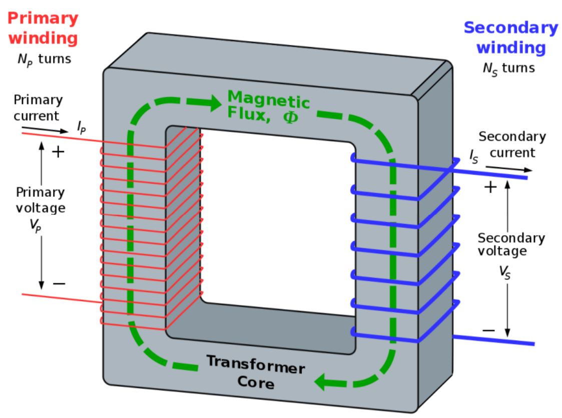
$$L(d) = K_f N^2 d\mu_v + K_f N^2 (1-d)\mu_0 = K_f N^2 \mu_{eff}(d)$$

Onde d é o deslocamento normalizado e sem unidades ($0 \leq d \leq 1$) e de onde tiramos que

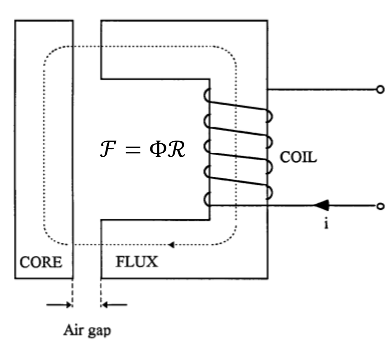
$$\mu_{eff}(d) = L(\mu_v - \mu_0) \cdot d + \mu_0$$

7

Sensores indutivos



https://en.wikipedia.org/wiki/Transformer



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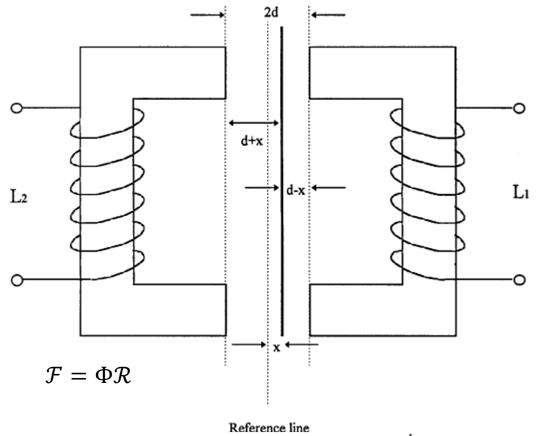
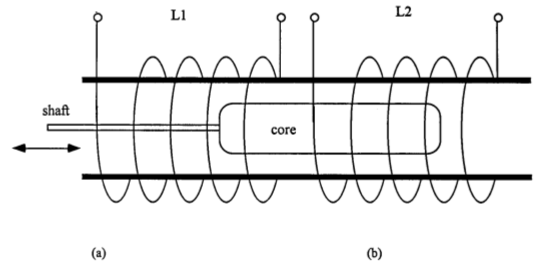
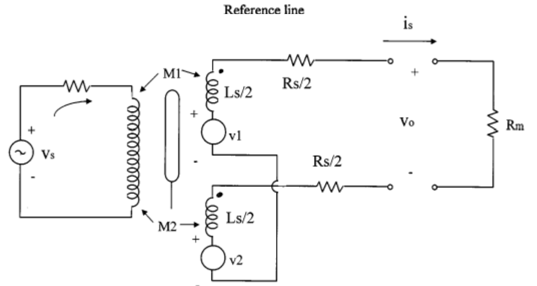
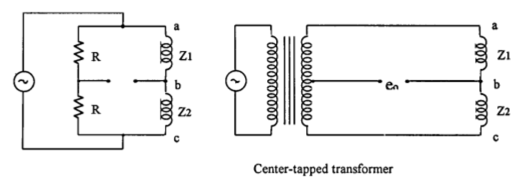


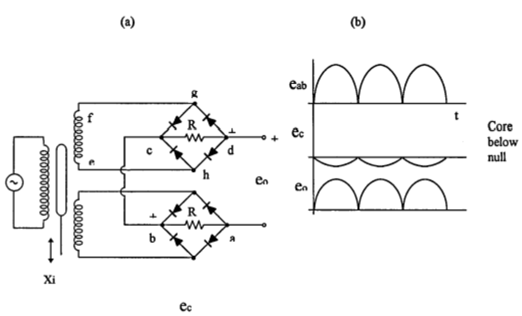
$$\mathcal{F} = \Phi\mathcal{R}$$







9



Desmodulador (para posicionamento)

10

Termistores

$$\beta = \frac{1}{R(\theta)}\frac{dR}{d\theta}$$

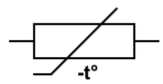
$$\mathrm{R}(\theta) = R_0 e^{\beta\left(\frac{1}{\theta}-\frac{1}{\theta_0}\right)}$$

$\beta \in [3000{,}5000]K$

Steinhart-Hart ±0.15 °C over the range of -50 to +150 °C

$$\frac{1}{T} = a + b \cdot ln(R) + c \cdot ln^3(R)$$

$T(K)$ -Temperatura Absoluta



NTC thermistor (IEC standard) https://eepower.com/resistor-guide/resistor-types/ntc-thermistor/#

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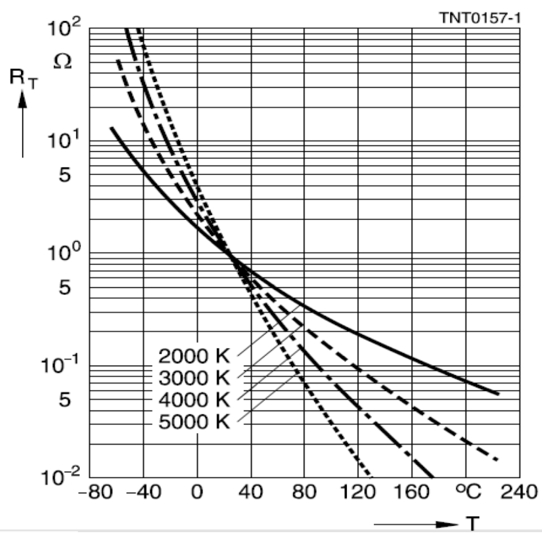
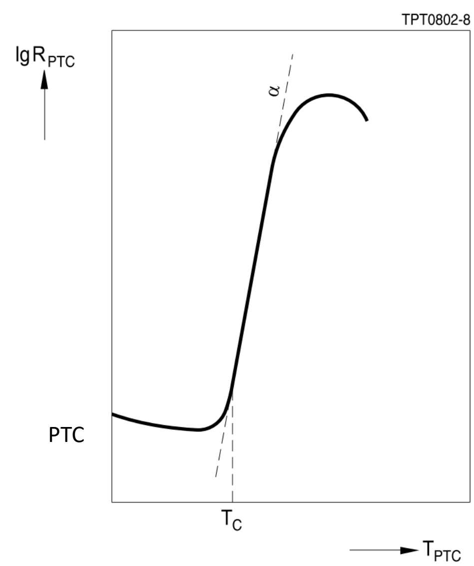
Termístores



NTC



PTC

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Termístores

100	19.957	29.935	38.640	55.905	0.091	0.082		
105	17.541	26.312	33.790	46.888	0.080	0.072		
110	15.453	23.180	29.664	42.918	0.070	0.063	0.105	0.076
115	13.663	20.494	26.123	37.795	0.062	0.056	0.093	0.067
120	12.114	18.171	23.091	33.409	0.055	0.049	0.083	0.058
125	10.778	16.168	20.472	29.618	0.049	0.044	0.074	0.051

Part Number	NCP□□XM222	NCP□□XW222	NCP□□XM332	NCP□□XW332	NCP□□XM472	NCP□□XW472	NCP□□XH682	NCP□□XW682
Resistance	2.2kΩ	2.2kΩ	3.3kΩ	3.3kΩ	4.7kΩ	4.7kΩ	6.8kΩ	6.8kΩ
B-Constant	3500K	3950K	3500K	3950K	3500K	3950K	3380K	3950K
Temp. (°C)	Resistance (kΩ)	Resistance (kΩ)	Resistance (kΩ)	Resistance (kΩ)	Resistance (kΩ)	Resistance (kΩ)	Resistance (kΩ)	Resistance (kΩ)
-40	46.786	75.961	70.179	113.941	105.705	162.279	133.043	234.787
-35	35.530	54.520	53.295	81.779	79.126	116.474	100.756	168.515
-30	27.162	39.607	40.743	59.411	59.794	84.615	77.076	122.422
-25	20.907	29.103	31.360	43.654	45.630	62.173	59.540	89.953
-20	16.203	21.601	24.304	32.401	35.144	46.147	46.401	66.766
-15	12.644	16.198	18.966	24.297	27.303	34.604	36.482	50.066
-10	9.934	12.264	14.901	18.396	21.377	26.200	28.904	37.906
-5	7.858	9.370	11.787	14.055	16.869	20.018	23.047	28.963
0	6.257	7.219	9.386	10.829	13.411	15.423	18.509	22.313
5	5.015	5.609	7.523	8.414	10.735	11.984	14.974	17.338
10	4.045	4.391	6.067	6.586	8.653	9.380	12.189	13.571
15	3.283	3.463	4.924	5.195	7.018	7.399	9.978	10.705
20	2.680	2.751	4.019	4.126	5.726	5.877	8.215	8.503
25	2.200	2.200	3.300	3.300	4.700	4.700	6.800	6.800
30	1.816	1.771	2.724	2.656	3.879	3.783	5.654	5.474
35	1.507	1.434	2.260	2.152	3.219	3.064	4.725	4.434
40	1.257	1.169	1.885	1.753	2.685	2.497	3.967	3.613
45	1.053	0.958	1.580	1.437	2.250	2.046	3.344	2.961
50	0.887	0.789	1.331	1.184	1.895	1.686	2.829	2.440
55	0.751	0.654	1.126	0.981	1.604	1.397	2.404	2.022
60	0.638	0.545	0.957	0.817	1.363	1.164	2.050	1.683
65	0.544	0.456	0.816	0.684	1.163	0.974	1.759	1.409
70	0.466	0.383	0.700	0.575	0.996	0.819	1.515	1.185
75	0.401	0.324	0.602	0.486	0.857	0.692	1.309	1.001
80	0.346	0.275	0.520	0.412	0.740	0.587	1.135	0.849
85	0.300	0.234	0.450	0.351	0.641	0.500	0.988	0.724
90	0.261	0.200	0.392	0.301	0.558	0.428	0.862	0.620
95	0.228	0.172	0.342	0.258	0.487	0.368	0.755	0.532
100	0.200	0.149	0.299	0.223	0.426	0.318	0.662	0.459
105	0.175	0.129	0.263	0.193	0.375	0.275	0.583	0.398
110	0.155	0.112	0.232	0.168	0.330	0.239	0.515	0.346
115	0.137	0.098	0.205	0.146	0.292	0.208	0.457	0.302
120	0.121	0.085	0.182	0.128	0.259	0.182	0.406	0.264
125	0.108	0.075	0.162	0.113	0.230	0.160	0.361	0.232

Continued on the following page.↗

Wheatstone Bridge (4-wired)

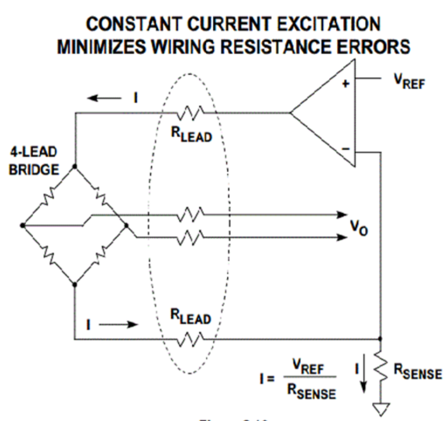

Figure 2.16

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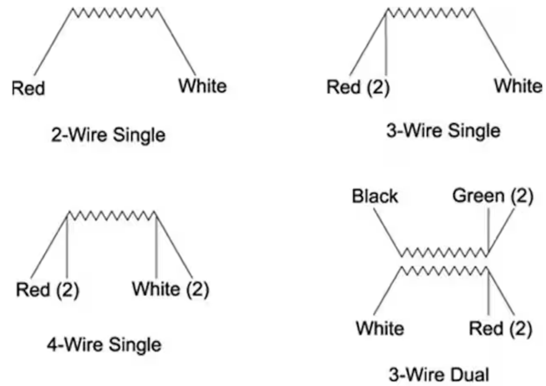
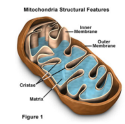
RTD

$\varrho = 10{,}6 \cdot 10^{-8}\Omega \cdot m$(Pt)
$\varrho = 1{,}77 \cdot 10^{-8}\Omega \cdot m$(Cu)
$\varrho = 9{,}71 \cdot 10^{-8}\Omega \cdot m$(Fe)
$\varrho = 2{,}65 \cdot 10^{-8}\Omega \cdot m$(Al)
$\varrho = 5{,}0 \cdot 10^{-7}\Omega \cdot m$(Constantin)

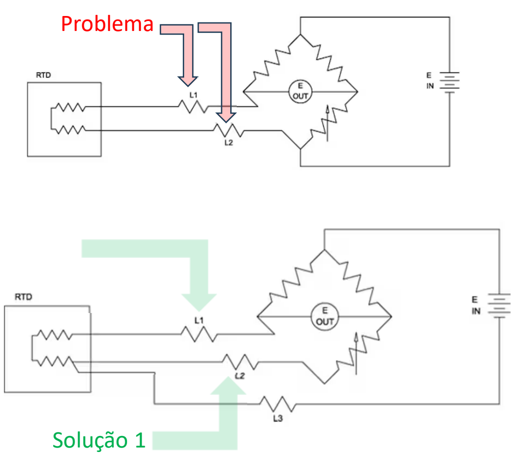

copper, silver, aluminium, platinum

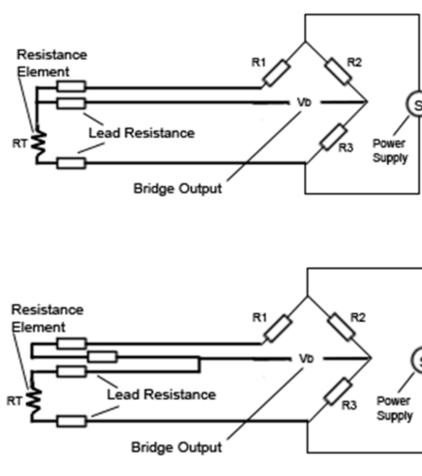


ϕ~ 0.5 to 3 µm

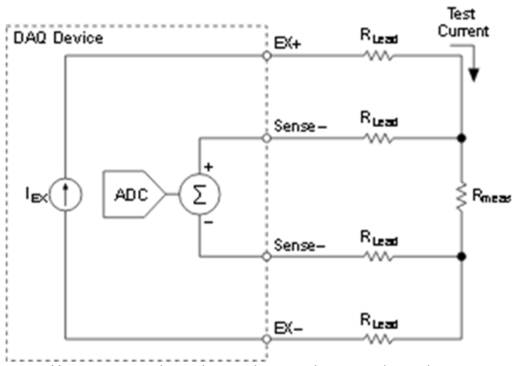


RTD





RTD 4-wire (mas não com Wheatstone bridge)



https://www.ni.com/docs/en-US/bundle/daqhelp/page/4wireres.html

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RTD

Calendar-Van Dusen

$$R_T = \begin{cases} R_0\left(1 + AT + BT^2 + CT^3(T-100)\right) \Leftarrow T < 0°C \\ R_0(1 + AT + BT^2) \Leftarrow T \geq 0°C \end{cases}$$

Simplificando

Coeficiente de Temperatura (T_0=0 °C) tb chamado 'α'

$$R_T = R_0(1 + AT)$$

ou

$$R_T = R_0\left(1 + \alpha(T - T_0)\right)$$

Valores típicos:

$$R_0 = 100\Omega \quad A = 0.00385°C^{-1}$$

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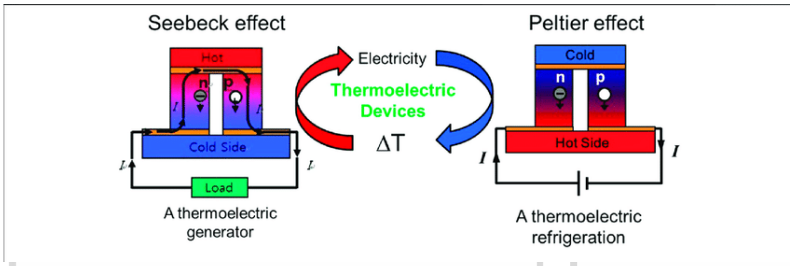
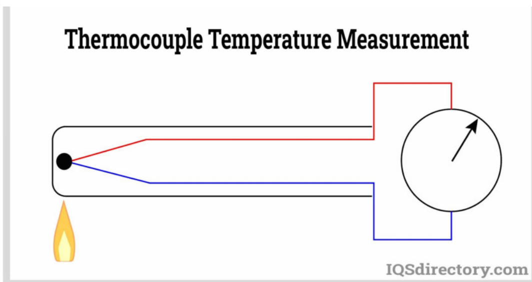
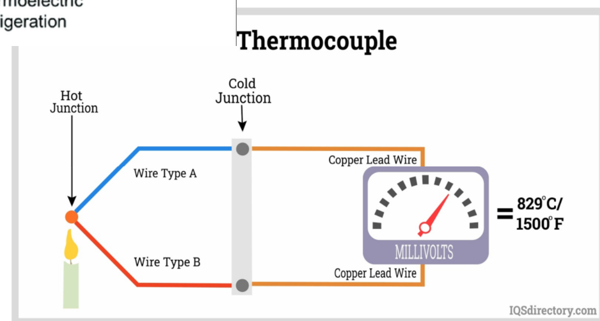
Termopares







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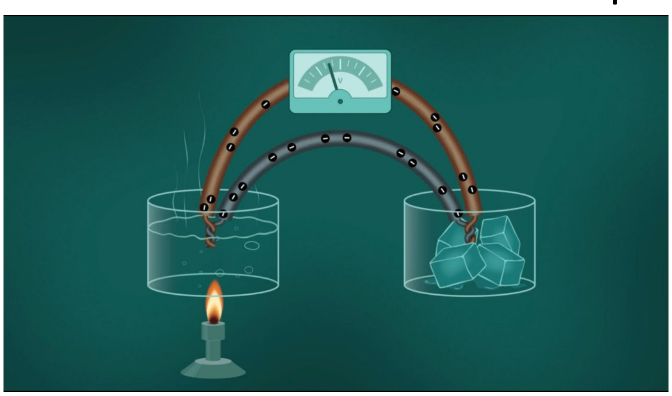
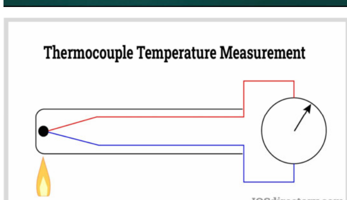
Termopares



TYPE K: CHROMEL-ALUMEL CJ=0°C

	0	5	10	15	20	25	30	35	40	45
−150	−4.81	−4.92	−5.03	−5.14	−5.24	−5.34	−5.43	−5.52	−5.60	−5.68
−100	−3.49	−3.64	−3.78	−3.92	−4.06	−4.19	−4.32	−4.45	−4.58	−4.70
−50	−1.86	−2.03	−2.20	−2.37	−2.54	−2.71	−2.87	−3.03	−3.19	−3.34
−0	0.00	−0.19	−0.39	−0.58	−0.77	−0.95	−1.14	−1.32	−1.50	−1.68
+0	0.00	0.20	0.40	0.60	0.80	1.00	1.20	1.40	1.61	1.81
50	2.02	2.23	2.43	2.64	2.85	3.05	3.26	3.47	3.68	3.89
100	4.10	4.31	4.51	4.72	4.92	5.13	5.33	5.53	5.73	5.93
150	6.13	6.33	6.53	6.73	6.93	7.13	7.33	7.53	7.73	7.93
200	8.13	8.33	8.54	8.74	8.94	9.14	9.34	9.54	9.75	9.95
250	10.16	10.36	10.57	10.77	10.98	11.18	11.39	11.59	11.80	12.01
300	12.21	12.42	12.63	12.83	13.04	13.25	13.46	13.67	13.88	14.09
350	14.29	14.50	14.71	14.92	15.13	15.34	15.55	15.76	15.98	16.19
400	16.40	16.61	16.82	17.03	17.24	17.46	17.67	17.88	18.09	18.30
450	18.51	18.73	18.94	19.15	19.36	19.58	19.79	20.01	20.22	20.43
500	20.65	20.86	21.07	21.28	21.50	21.71	21.92	22.14	22.35	22.56
550	22.78	22.99	23.20	23.42	23.63	23.84	24.06	24.27	24.49	24.70
600	24.91	25.12	25.34	25.55	25.76.	25.98	26.19	26.40	26.61	26.82
650	27.03	27.24	27.45	27.66	27.87	28.08	28.29	28.50	28.72	28.93
700	29.14	29.35	29.56	29.77	29.97	30.18	30.39	30.60	30.81	31.02
750	31.23	31.44	31.65	31.85	32.06	32.27	32.48	32.68	32.89	33.09
800	33.30	33.50	33.71	33.91	34.12	34.32	34.53	34.73	34.93	35.14
850	35.34	35.54	35.75	35.95	36.15	36.35	36.55	36.76	39.96	37.16
900	37.36	37.56	37.76	37.96	38.16	38.36	38.56	38.76	38.95	39.15
950	39.35	39.55	39.75	39.94	40.14	40.34	40.53	40.73	40.92	41.12
1000	41.31	41.51	41.70	41.90	42.09	42.29	42.48	42.67	42.87	43.06
1050	43.25	43.44	43.63	43.83	44.02	44.21	44.40	44.59	44.78	44.97
1100	45.16	45.35	45.54	45.73	45.92	46.11	46.29	46.48	46.67	46.85
1150	47.04	47.23	47.41	47.60	47.78	47.97	48.15	48.34	48.52	48.70
1200	48.89	49.07	49.25	49.43	49.62	49.80	49.98	50.16	50.34	50.52
1250	50.69	50.87	51.05	51.23	51.41	51.58	51.76	51.94	52.11	52.29
1300	52.46	52.64	52.81	52.99	53.16	53.34	53.51	53.68	53.85	54.03
1350	54.20	54.37	54.54	54.71	54.88					



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Termopares

Thermocouple: Type J, K, E, T, N, B, S, R;

Type J (Iron/Constantan) Is Well Suited To Oxidizing Atmospheres

Type T Thermocouple (Copper/Constantan)

Type E Thermocouple (Nickel-Chromium/Constantan)

International Thermocouple Color Codes - Thermocouple and Extension Grade Wires

THERMO-COUPLE TYPE	U.S. & CANADIAN (ANSI/MC96.1, ANSI/ASTM E230)				International	International	Czech British	Netherlands German	Japanese	French
	ALLOY COMBINATION	THERMOCOUPLE GRADE	EXTENSION GRADE	PLUG & JACK	IEC 584-3	IEC 584-3 Intrinsically Safe	BS 1843	DIN 43710	JIS C 1610	NFC 42-324
T	Copper / Constantan (Copper-Nickel)	Brown; + Blue; - Red	Blue; + Blue; - Red	Blue	Brown; + Brown; - White	Blue; + Brown; - White	Blue; + White; - Blue	Brown; + Red; - Brown	Brown; + Red; - White	Blue; + Yellow; - Blue
J	Iron (magnetic) / Constantan (Copper-Nickel)	Brown; + White; - Red	Black; + White; - Red	Black	Black; + Black; - White	Blue; + Black; - White	Black; + Yellow; - Blue	Blue; + Red; - Blue	Yellow; + Red; - White	Black; + Yellow; - Black
E	Nickel - Chromium / Constantan (Copper- Nickel)	Brown; + Purple; - Red	Purple; + Purple; - Red	Purple	Purple; + Purple; - White	Blue; + Purple; - White	Brown; + Brown; - Blue	Black; + Red; - Black	Purple; + Red; - White	Purple; + Yellow; - Purple
K	Nickel - Chromium / Nickel - Aluminium (magnetic)	Brown; + Yellow; - Red	Yellow; + Yellow; - Red	Yellow	Green; + Green; - White	Blue; + Green; - White	Red; + Brown; - Blue	Green; + Red; - Green	Blue; + Red; - White	Yellow; + Yellow; - Purple
N	Nicrosil (Nickel-Chromium- Silicon) / Nisil (Nickel-Silicon-Magnesium)	Brown; + Orange; - Red	Orange; + Orange; - Red	Orange	Pink; + Pink; - White	Blue; + Pink; - White	Orange; + Orange; - Blue	No Standard (Use American Color Codes)	No Standard (Use American Color Codes)	No Standard (Use American Color Codes)
S	Platinum Rhodium -10% / Platinum	None Established	Green; + Black; - Red	Green	Orange; + Orange; - White	Blue; + Orange; - White	Green; + White; - Blue	White; + Red; - White	Black; + Red; - White	Green; + Yellow; - Green
R	Platinum Rhodium -13% / Platinum	None Established	Green; + Black; - Red	Green	Orange; + Orange; - White	Blue; + Orange; - White	Green; + White; - Blue	White; + Red; - White	Black; + Red; - White	Green; + Yellow; - Green
B	Platinum Rhodium - 30% / Platinum Rhodium - 6%	None Established	Gray; + Gray; - Red (Compensated Cable)	White (Uncom-pensated)	Gray; + Gray; - White		No Standard (Use Copper Wire)	Gray; + Red; - Gray (Compensated Cable)	Gray; + Red; - Gray (Compensated Cable)	No Standard (Use Copper Wire)
C	Tungsten Rhenium - 5% / Tungsten Rhenium - 26%	None Established	Red; + Green; - Red	Red				No Standard (Use American Color Codes)	No Standard (Use American Color Codes)	No Standard (Use American Color Codes)

Type R Thermocouple (Platinum Rhodium -13% / Platinum)

Type B Thermocouple (Platinum Rhodium – 30% / Platinum Rhodium – 6%)

Type S Thermocouple (Platinum Rhodium - 10% / Platinum)

Type C Thermocouple (Tungsten Rhenium - 5% / Tungsten Rhenium - 26%)

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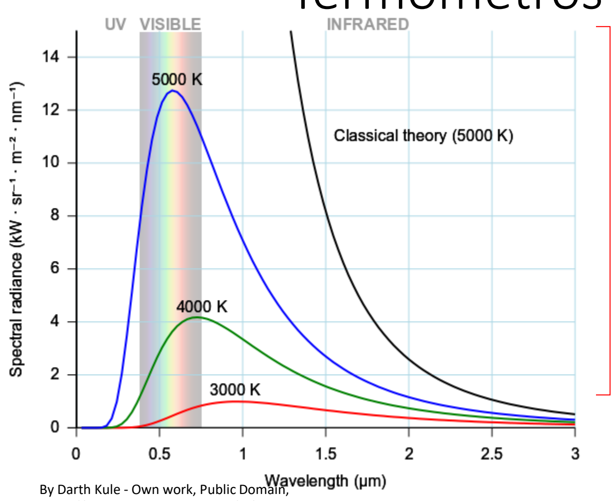
Termómetros de Radiação



By Darth Kule - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=10555337

Lei de Stephan-Boltzmann

$$R_T = \int_0^{+\infty} R_{b\lambda}(\theta,\lambda)\cdot d\lambda = \sigma\theta^4 (W\cdot m^{-2})$$
$$\sigma = 5{,}67\times 10^{-8} W\cdot m^{-2}\cdot K^{-4}$$

Lei de Wien

$$\lambda_{Máx} = \frac{2898}{\theta}(\mu m)$$

Lei de Planck

$$\begin{cases} R_{b\nu}(T,\nu) = \dfrac{2h\nu^3}{c^2}\dfrac{1}{e^{h\nu/kT}-1} \\ R_{b\lambda}(T,\lambda) = \dfrac{2hc^2}{\lambda^5}\dfrac{1}{e^{hc/\lambda kT}-1} \end{cases}$$

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Algumas Aplicações

https://www.researchgate.net/publication/342486276_The_Influence_of_Biomechanical_Stability_on_Bone_Healing_and_Fracture-Related_Infection_The_Legacy_of_Stephan_Perren/figures

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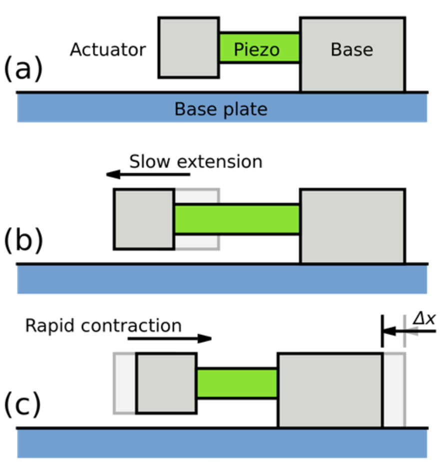
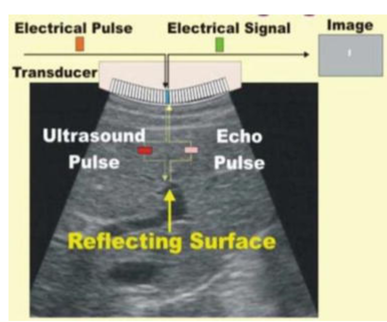
Algumas Aplicações

https://en.wikipedia.org/wiki/Piezoelectric_motor





https://www.intechopen.com/chapters/77225

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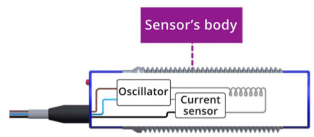
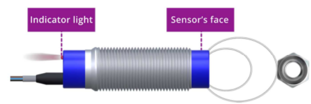
Algumas Aplicações

https://www.ifm.com/pt/pt/product/MGS201?source=gs&gad_source=1&gclid=CjwKCAjwkuqvBhAQEiwA65XxQOefvx8xC8oupk5DN_rcWMoXege9IhhsexZR5ZJ1yCFXQ682cf70vBoCCaoQAvD_BwE



https://www.realpars.com/blog/inductive-sensor REALPARS



REALPARS

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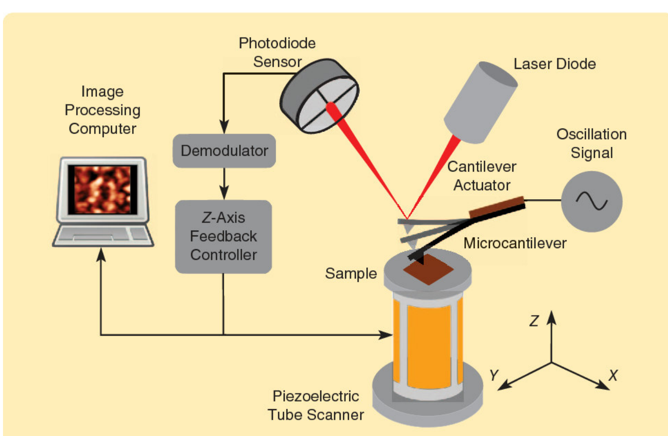
Algumas Aplicações



https://www.semanticscholar.org/paper/Control-Techniques-for-Increasing-the-Scan-Speed-in-Fairbairn-Moheimani/649c956c8fdd15eda7febf17ca9caad52a08fe12

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Algumas Aplicações

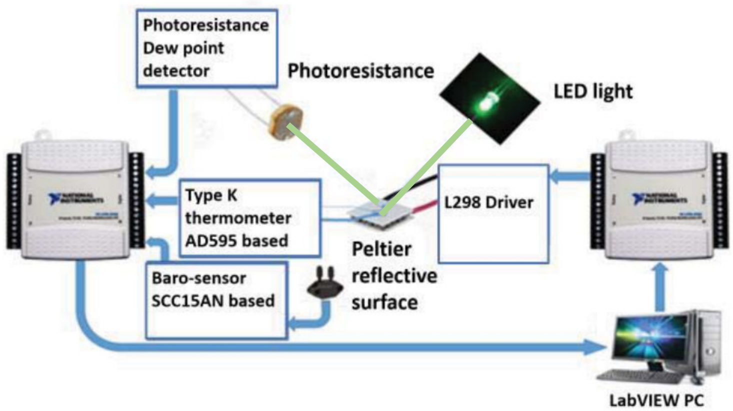

Fig. 1. Functional layout of the dew point humidity sensor.
https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8593483

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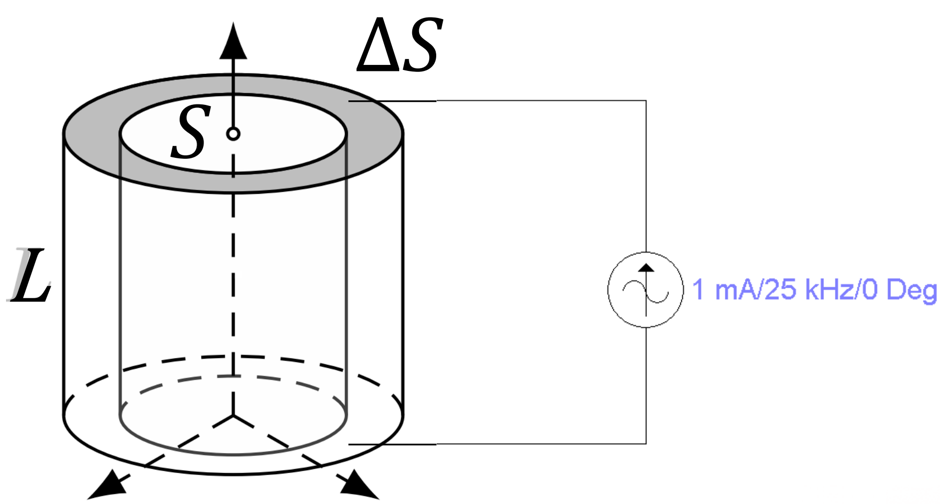
Algumas Aplicações



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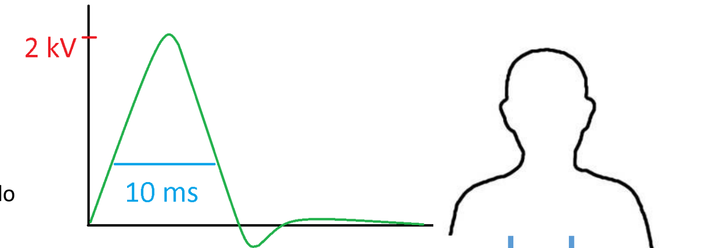
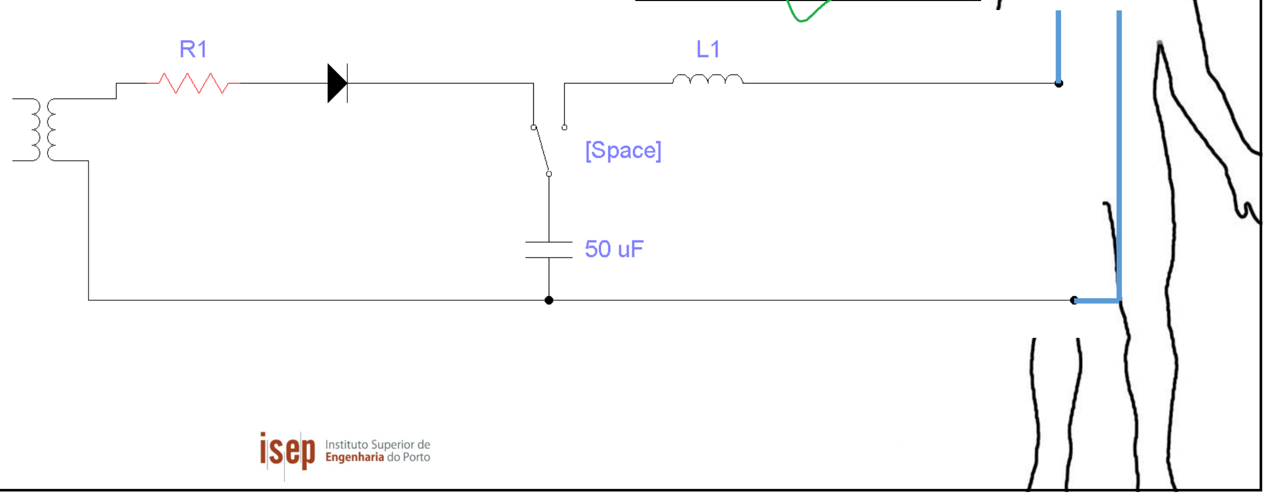
Algumas Aplicações

Um Desfibrilador muito simplificado





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